ChatGPT

Global Precedents

Several countries have experimented with real-time video monitoring of polling places. In **India**, for example, the state of Kerala (2026) deployed AI-enabled PTZ CCTV cameras to *count heads and monitor queue lengths* at polling booths. Computer-vision models (likely based on CNN detectors like YOLO) count people and flag anomalies (e.g. sudden crowding) in ~2 seconds [1]. Officials in Kerala report that alerts (with booth ID and wait estimate) are validated by human analysts and used to manage queues and dispatch officials. West Bengal (2026) similarly installed cameras inside/outside each of ~80,000 booths, feeding a central control-room ("war room") with live video. A team of "micro-observers" (≈200 people) review streams, aided by AI for anomaly alerts [2].

Beyond India, **commercial platforms** exist. *VMukti* (India) sells "booth-level security" systems using CV to detect crowds, weapons, etc. *Revisor* (PollWatcher.ai) is an AI software for election oversight: it "accurately monitors physical objects, identifies voting events" and claims up to 98% accuracy in counting voters and flagging anomalies [3]. Academic research likewise shows CV can spot fraud-like patterns: e.g. IDEA reports that CCTV+AI could flag "if a person appears twice at the same polling site or at multiple locations" [4], and generally detect "incidents or anomalies" (changes from normal frame patterns) [5].

Empirical results are still limited, but initial reports suggest these systems can **significantly raise alerts** (e.g. long queues, duplicates) with fairly high accuracy (vendor claims ~98% for turnout counting [3]). Failures arise from occlusions (people block view), poor image quality, and false alarms from irrelevant motion (e.g. swaying trees) [1]. Critics note that cameras only show *certain* violations (crowds, overt coercion) and cannot capture hidden fraud like ballot-box stuffing [6]. They also warn of **cost and privacy issues** [7] [4]. In sum, past precedents (Kerala, West Bengal, etc.) demonstrate basic viability but also highlight the need for human oversight of AI flags.

Figure: CCTV camera and queue at an Indian polling station. Modern deployments (e.g. Kerala 2026) use PTZ cameras + AI to count voters and monitor queues [1].

Alignment with Armenian Electoral Code

Armenia's law already anticipates video monitoring. **Article 8** (added 2016) **requires** fixed cameras at every precinct recording *"the voting process and the process of summarisation"* and webcast in real time [8]. (These cameras must cover the ballot box, station entrance, and up to 50 m radius [8].) At the same time **Article 6** guarantees *secret ballot*: "Voting shall be held by secret ballot… Control over the free expression of will of an elector shall be prohibited" [9]. In practice this means the camera view must not reveal how a voter marks the ballot. A well-designed CV system can respect this (e.g. camera angles that see people but not ballot choices).

Key CV capabilities can map to specific violations in the code:

- **Unauthorized persons in the booth:** Armenian law (v.2018) states that an assisted voter may invite *exactly one* helper into the voting booth, and *only one voter at a time* may be assisted [10]. Any other person entering the booth is prohibited [11]. Computer vision models (person detection + tracking/ReID) can flag if more than two people occupy the booth or if the same helper appears assisting multiple voters. For example, if tracking shows a non-disabled helper entering a booth repeatedly, that violates the rule.

- **One-at-a-time voting:** By law voters must enter and vote one-by-one (no cluster voting) [12]. CV-based people-counting and flow analysis can detect if multiple voters enter or linger in the voting room simultaneously. For instance, a sudden build-up of more than one person in the booth area can trigger an alert. (The code actually limits total persons in the voting room at once – one session allows up to ~15 at a time, and requires one-by-one admission [13] [14].)

- **Distance restrictions:** Candidates and party agents are prohibited from loitering in or near polling centers (within 50 m) during voting [15]. In principle, CV (with camera-perimeter views) could detect familiar individuals (via face or clothing recognition) entering that forbidden zone, or simply large groups gathering near the entrance. Likewise, CV could watch for illicit filming: if a smartphone is raised toward a ballot booth, pose-estimation might flag "person holding phone toward booth," which could indicate a breach of secrecy.

- **Crowd management:** While not all crowding is illegal (observers/media are exempt), the Electoral Code emphasizes smooth orderly flow. CV can enforce distance markers or maximum room occupancy and alert staff to unusually long queues or crowd build-ups that might intimidate voters.

In summary, CV could directly address many RA Code concerns: secret-ballot compliance (by enforcing booth privacy), improper assistance (flagging extra helpers or repeats), and crowd violations (monitoring head counts and movement). All such uses would need to respect Article 6's secrecy duty [9] (i.e. not identify actual votes or pressure individuals). Importantly, Armenian law already mandates video monitoring [8], so the legal framework would permit – and even facilitate – an AI add-on to analyze those feeds for rule violations.

Technical Architecture & Constraints

Ingestion: 1800 RTSP/HLS streams must be handled in parallel. A hybrid architecture is optimal: lightweight *edge devices* at or near polling centers (e.g. local NVRs or Jetson-class units) can preprocess video (motion detection, downsampling) and flag candidate anomalies, while a central cloud/Kubernetes cluster takes higher-level analysis. Video can be streamed (or key frames) via secured links to data centers. Solutions like Kafka or GStreamer pipelines could feed frames to inference services.

CV Models: We recommend modern real-time detectors and trackers. For person detection, YOLOv8 or YOLOv9 (or comparable one-stage models) offer a good speed/accuracy trade-off. For multi-object tracking, **DeepSORT** or **ByteTrack** can maintain identity across frames, allowing person Re-identification (ReID). ReID models (e.g. OSNet) can recognize when the same person returns. Pose-estimation models (e.g. OpenPose or MediaPipe) could identify actions (like a voter filling a ballot or holding a phone). Facial recognition is technically possible but legally sensitive (especially under “no control” of vote), so a safer approach is to use non-biometric ReID (clothing/appearance) when needed.

Low-Quality Camera Handling: Many polling cameras will be low-res or at odd angles. Models must be fine-tuned or trained on similar data. Data augmentation (angles, occlusions) and lightweight architectures (like YOLO-NAS) help. Because angles may not see faces clearly, algorithms must rely on body detection and tracking rather than face recognition. For distance measurement, one can calibrate camera FOV to estimate crowd density or use floor markers (if visible).

Infrastructure & Latency: To analyze 1800 streams at ~5–15 fps (real time), substantial compute is needed. For example, a single GPU (NVIDIA A100/H100) can handle roughly 20–30 concurrent low-res streams on YOLO inference. Thus we may need on the order of 50–100 GPUs (or dozens of multi-GPU servers) for full parallelism, or distribute more workload to edge GPUs. Alternatively, using specialized hardware (e.g. NVIDIA Jetson Orin X at each station) can filter obvious frames locally and send only metadata/alerts upstream, greatly reducing bandwidth. The central system would then aggregate alerts in real time. End-to-end latency (capture→detect→alert) should target under 1–2 seconds to be useful; this is achievable with optimized inference (e.g. TensorRT) and a well-provisioned network.

Bottlenecks: The biggest constraints are network bandwidth (thousands of video feeds) and compute cost. Strategies: process at lower frame rates or resolution; detect “events” (e.g. significant motion) before full analysis; use CPU-only trackers after initial detection. For cloud deployment, we might use auto-scaling GPU clusters and microservices. On-premise (perhaps in Yerevan) might be chosen for security.

Occlusions: People blocking the camera pose a challenge. Mitigations include: installing **multiple cameras** per booth (e.g. one angled at the queue and one inside at a fixed vantage), so if one is occluded the other might still see the voter. The AI can also use temporal logic: if the ballot box is occluded mid-voting but a tracked person entered and later exited, we infer a voting event happened. Advanced approaches (3D reconstruction from multiple views, or using overhead cameras) are possible but complex. In practice, strong system design should emphasize clear sightlines, since a fully blind spot is hard to compensate.

AI vs. Human Observers (Cost & Synergy)

Cost-Effectiveness: Staffing 1800 full-time observers (one per station) is extremely costly and logistically daunting. At even \$50 per observer-day, that's \$90k per day (plus travel/training) – and human attention is limited. An AI system has high upfront cost (hardware, development, legal compliance) but then minimal per-day expense (power, maintenance). It can run continuously (24/7) without fatigue. AI will generate many alerts (and some false positives), so a lean human review team can focus only on flagged incidents. For example, in Kerala each AI alert was reviewed by 30–40 analysts to confirm validity [1]. This "AI-as-sensor, human-as-expert" model scales far beyond one-observer-per-booth.

Effectiveness: AI cannot replace all nuances of human judgment. It may mislabel benign events (e.g. flagging a party member handing water to a voter as "crowding") or miss subtle pressure. Conversely, AI excels at always-on surveillance: it can count people in/out, detect duplicate entries, or identify repeat helper visits automatically. A hybrid strategy is best: the AI acts as a real-time **alert system**, continuously scanning all booths for anomalies (long queues, multiple entries by one person, extra faces in a booth). A central rapid-response unit of legal/election experts can then triage these alerts. For instance, if AI flags "same man entered booth twice in 5 minutes" at station #123, the team can quickly dispatch a local monitor or instruct police to verify. This synergy was anticipated by election experts: AI "flags can be followed up by EMB officials or observers" [4]. In practice, Armenia could equip a mobile "tiger team" to accompany the AI, reviewing video clips or visiting polling stations after an alert.

Hardware Limits & Feasibility: The concept is *technically feasible* (building on precedents in India) but challenging. Hardware must be deployed at scale (GPU clusters or edge boxes) and robust networking ensured at every station. There is risk of outages or covered cameras (incidentally or maliciously). In terms of election law, Armenia *explicitly allows* video recording (Article 8) but still requires vote secrecy (Art.6). Our CV system must be designed to **detect violations without revealing ballots**; in fact, it often enforces the secrecy rule (by ensuring no one illicitly observes a vote).

Realistic Assessment: An 1800-station AI vision system is *ambitious* but not impossible. India's experience shows it can be done at a national scale, though with heavy investment. For Armenia's National Assembly election, cameras are already mandated – adding AI analysis is a logical extension. However, success depends on careful planning: quality of camera installation, choice of efficient models (e.g. YOLOv8/9 for detection, DeepSORT for tracking, a simple ReID or even just 2D tracking), and extensive testing against occlusion scenarios. The system would not catch every infraction, but could significantly amplify the reach of limited human observers. Given the technological and budgetary constraints, a phased approach (pilots in high-risk areas, incremental rollout) is prudent. If well-executed, AI vision can serve as an *early-warning network*, making Armenian elections safer by swiftly drawing attention to procedural breaches, while still relying on human judgment to act on those alerts.

Sources: Kerala/India deployments [1] [2]; VMukti/Revisor product info [3]; IDEA survey of AI uses [5]; Armenian Electoral Code (v.2016/18) texts [9] [8] [10]; expert critique on cameras [6] [7].

---

[1] Kerala Elections Test AI Public Surveillance at Polling Queues - AI CERTs News
https://www.aicerts.ai/news/kerala-elections-test-ai-public-surveillance-at-polling-queues/

[2] AI-powered control room with 90 screens to monitor 80,700 polling booths 24x7 during Bengal elections
https://www.newindianexpress.com/elections/west-bengal-elections/2026/Apr/08/ai-powered-control-room-with-90-screens-to-monitor-80700-polling-booths-24x7-during-bengal-elections

[3] Revisor - AI Tool for Elections Analysis | AIAXIO
https://aiaxio.com/tools/ai/revisor/?src=search

[4] [5] Artificial Intelligence for Electoral Management
https://www.idea.int/sites/default/files/2024-04/artificial-intelligence-for-electoral-management.pdf

[6] [7] Use of Cameras/Webcams in Polling Stations as Confidence Building Measures —
https://aceproject.org/electoral-advice/archive/questions/replies/291099047

[8] [9] [10] [11] [12] [13] [14] [15] Microsoft Word - RA_Electoral_Code_ENG_asof_20_10_2018_ENG.DOCX
https://www.translation-centre.am/pdf/Translat/HH_Codes/Electoral_Code_04052018_en.pdf